

EX. NO: 5

IMPLEMENTATION OF RAIL FENCE – ROW & COLUMN TRANSFORMATION TECHNIQUE

AIM:

To write a C program to implement the rail fence transposition technique.

ALGORITHM:

STEP-1: Read the Plain text.

STEP-2: Arrange the plain text in row columnar matrix format.

STEP-3: Now read the keyword depending on the number of columns of the plain text.

STEP-4: Arrange the characters of the keyword in sorted order and the corresponding columns of the plain text.

STEP-5: Read the characters row wise or column wise in the former order to get the cipher text.

PROGRAM:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```
    int i, j, k, l;
```

```
    char a[100], c[100], d[100];
```

```
    printf("\n\t\tRAIL FENCE CIPHER\n");
```

```
    printf("\nEnter the Plain Text: ");
```

```
    scanf("%s", a);
```

```
    l = strlen(a);
```

```

/* Encryption */

j = 0;

for (i = 0; i < l; i++)
{
    if (i % 2 == 0)
        c[j++] = a[i];
}

for (i = 0; i < l; i++)
{
    if (i % 2 != 0)
        c[j++] = a[i];
}

c[j] = '\0';

printf("\nEncrypted Text : %s\n", c);

/* Decryption */

if (l % 2 == 0)
    k = l / 2;
else
    k = (l / 2) + 1;

j = 0;
for (i = 0; i < k; i++)
{
    d[j] = c[i];
}

```

```
        j += 2;
    }

    j = 1;
    for (i = k; i < l; i++)
    {
        d[j] = c[i];
        j += 2;
    }

    d[l] = '\0';

    printf("Decrypted Text : %s\n", d);

    return 0;
}
```

OUTPUT:



The screenshot shows a code editor interface with a dark theme. The top toolbar includes buttons for Run, Debug, Stop, Share, Save, Beautify, and a download icon. Below the toolbar, the console output is displayed in a monospaced font. The text in the console reads: "RAIL FENCE CIPHER", "Enter the Plain Text: KESH", "Encrypted Text : KSEH", "Decrypted Text : KESH", "...Program finished with exit code 0", and "Press ENTER to exit console." followed by a cursor icon.

```
RAIL FENCE CIPHER

Enter the Plain Text: KESH

Encrypted Text : KSEH
Decrypted Text : KESH

...Program finished with exit code 0
Press ENTER to exit console.
```

RESULT:

Thus the rail fence algorithm had been executed successfully